

Soroush Dianaty

Physician-Scientist & PhD Student | Biomedical Informatics & Data Science

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As a health informatician with a background in medicine, I bring clinical domain expertise to health wearables and biosignal analytics in cardiology, with a focus on cardio-oncology. My work centers on anchoring wearable data in physiological rationale and cardiovascular modeling—translating continuous sensor signals into clinically meaningful biomarkers while developing evaluation frameworks for trustworthy clinical AI.

Education

PhD, Biomedical Informatics and Data Science Aug 2025 – Present

Arizona State University

Doctor of Medicine (M.D.) Sep 2016 – Jun 2024

Tehran Medical Sciences Branch, Islamic Azad University (IAUTMU)

Thesis: Cost-effectiveness of plasmapheresis vs hemoperfusion in critically ill COVID-19 patients

Skills

Machine Learning & AI:

- Deep Learning & Classical ML
- LLM Fine-Tuning (LoRA, PEFT)
- Prompt & Context Engineering
- LLM Application Development

Programming & Data:

- Python
- SQL
- SAS
- Linux & Git

Health Data & Interoperability:

- FHIR
- SNOMED-CT, LOINC, ICD
- EHR Systems (Epic)

Research Methods:

- Regression Modeling
- Survival Analysis
- Cost-effectiveness Analysis

Publications

Peer-reviewed Articles

1. Lee P, Dhadwal AS, Kaiser M, Dianaty S, Lott E, Singh G, Chern D, Walonoski JA, Grando A. "Assessing the Effectiveness and Scalability of Fast Healthcare Interoperability Resource-Based Granular Data Segmentation Technology". *Applied Clinical Informatics*. 2026. doi:10.1055/a-2863-4129
2. Dianaty S, Gholami F, Gholamrezaie HR, Mirzaei A. "Cost-effectiveness of plasmapheresis and hemoperfusion in COVID-19 survivors: A six-month follow-up analysis after hospital discharge". *Therapeutic Apheresis and Dialysis*. 2025. doi:10.1111/1744-9987.14235
3. Raiszadeh M, Khosronejad A, Dianaty S, Ghorbani Yekta B. "Novel repair of acute Stanford type B aortic dissection using combined endovascular graft and transfemoral replacement of vascular plug". *Catheterization and Cardiovascular Interventions*, 102(7), 1287–1290. 2023. doi:10.1002/ccd.30849
4. Dianaty S, Khodadadi S, Alimoghaddam R, Mirzaei A. "Comparison of outcomes and costs of extracorporeal blood purification therapies in critically ill COVID-19 patients". *Therapeutic Apheresis and Dialysis*, 27(3), 505–516. 2023. doi:10.1111/1744-9987.13948
5. Khoshgoftar Some Saraii Z, Dianaty S, Rouhollah F, Zare N, Ghorbani Yekta B. "Reproductive status of male rat offspring following exposure to methamphetamine during intrauterine life: An experimental study". *International Journal of Reproductive BioMedicine*, 21(2), 175–184. 2023. doi:10.18502/ijrm.v21i2.12809
6. Yaminifar L, Dianaty S, Shahverdi Z, Yaraghi M, Ghorbani Yekta B. "Differential diagnosis of benign ovarian cysts using tumor markers in serum and cyst fluid". *South East Asia Journal of Medical Sciences*, 5(2), 1–7. 2021.
7. Darabi R, Ghoreishi B, Dianaty S, Motevalli MS. "Stress and menstrual disorders among Iranian medical students: A cross-sectional study". *South East Asia Journal of Medical Sciences*, 5(2), 8–15. 2021.

Conference Abstracts & Proceedings

1. Dianaty S, Kaiser M, Murcko A, Grando A. "Early Evidence for Context-Aware Large Language Models (LLMs) in Sensitive Health Data Classification". In *AcademyHealth Annual Research Meeting 2026*, Seattle, WA. 2026 May 30 – Jun 2.
2. Dianaty S, Soumma SB. "Explainable early prediction of acute kidney injury using first 24-hour physiologic and clinical data". In *2nd Arizona Digital Health Symposium*, Phoenix, AZ. 2026 May 5.
3. Dianaty S, Ghorbani Yekta B. "Effects of Descurainia Sophia products on urinary and renal function". In *Congress of Invention, Patenting and Manufacturing of Experimental, Pharmaceutical and Clinical Products*, Tehran, Iran. 2018 Feb 25.

Book Chapters

- Chapters in Practical Guide to Psychiatric Medications (2nd edition). In *Practical Guide to Psychiatric Medications*, 2nd ed., MTP Psychiatry (2022).
Four chapters: Introduction to Digital Psychiatry; The COVID-19 Pandemic and Mental Health; Apps and Wearables for Mental Health; Future Advances in Mental Health.

Research Experience

Cardiology, Wearable Health & Biosignal Modeling (SHANDHI Lab) Aug 2026 – Present

Arizona State University · Funded by AHA · PI: Dr. M. M. H. Shandhi | Tempe, AZ

- Anchoring continuous wearable biosignals and sensor data to physiological rationale and cardiovascular modeling.
- Developing digital biomarker translation pipelines and evaluation frameworks for trustworthy clinical AI in cardio-oncology.

Substance Use Health Record Sharing (SHARES) Oct 2024 – Present

Arizona State University · Funded by NIDA · PI: Dr. Adela Grando | Tempe, AZ

- Developed and validated EHR segmentation rules to support automated substance-use-disorder record sharing.
- Applied clinical domain knowledge to improve the accuracy and interpretability of structured health data.

Development and Evaluation of an Online Decision Aid Nov 2024 – Present

Arizona State University · Funded by AHRQ · PI: Dr. Dongwen Wang | Tempe, AZ

- Developed patient-facing educational content and backend scoring logic for an online contraception decision aid tailored to transgender and gender-nonconforming users assigned female at birth.

Computational Medicine Center Aug 2023 – May 2024

K. N. Toosi University of Technology | Tehran, Iran

- Led interdisciplinary development of neural networks for cardiovascular disease detection from PPG signals.
- Built Python pipelines for data acquisition, preprocessing, and analysis of EHR-linked biosignal data.

Universal Scientific Education and Research Network (USERN) Mar 2022 – Mar 2023

Tehran University of Medical Sciences | Tehran, Iran

- Coordinated logistics and research-related content development for USERN events.
- Automated certificate generation with Python, reducing processing time by 60%.

Neuroscience Research Center May 2018 – Jan 2020

Department of Physiology, IAUTMU | Tehran, Iran

- Conducted data analysis and visualization in support of neuropharmacology research.

Teaching Experience

Teaching Associate Aug 2025 – Present

College of Health Solutions, Arizona State University

- Supported instruction for BMI 201 (Introduction to Clinical Informatics) and BMI 601 (Health Informatics) through assignment development, discussion facilitation, grading, and student guidance.

Online Instructor Sep 2023

Academic Writing & AI Workshop

- Designed and delivered a 6-hour workshop for 50+ attendees on the use of AI tools in academic writing.

Clinical Experience

Family Physician Oct 2024 – Jul 2025

Qeshlaq Rural Health Center | Abyek, Qazvin, Iran

- Delivered primary and preventive care for a rural population of 7,500+ across 17 villages and 5 satellite health centers.
- Managed chronic disease, prenatal, adolescent, and mental health care within Iran's national EHR system.
- Coordinated care with multidisciplinary teams across distributed rural service sites.

Clinical Rotations Oct 2020 – May 2024

IAUTMU Hospital Network | Tehran, Iran

- Completed 7,000+ hours of clinical rotations across 6 hospitals and 16 specialties.

Awards & Honors

First Place (\$2,500), Arizona BioChallenge — Nucleate Arizona

2025-10-01

Graduate Fellowship (\$4,845) — Graduate College, Arizona State University

2025-08-01

Active Member Award — Student Research Committee, IAUTMU

2018-10-01

Licenses & Certifications

Inter-Personal Skills in Engineering Project Management | Ira A. Fulton Schools of Engineering, Arizona State University

Jun 2026 – Aug 2026

Fundamentals of Engineering Project Management | Ira A. Fulton Schools of Engineering, Arizona State University

Jun 2026 – Aug 2026

Medical License (No. 222197) | Medical Council of Iran (IRIMC)

Sep 2024

IRB - Biomedical Research | CITI Program Course

Feb 2026

Human Subjects Research | CITI Program Course

Sep 2023

Responsible Conduct of Research & Data Privacy | CITI Program Course

Sep 2023

Professional Activities

Peer Review

- Frontiers in Pharmacology — 1 manuscript reviewed
- Italian Journal of Gynaecology & Obstetrics — 2 manuscripts reviewed

Memberships

- American Medical Informatics Association (AMIA) (Dec 2025 – Present)
- ASU CHS Council of Inclusive Excellence (CIE) (Oct 2025 – Present)
- USERN Executive Committee (Mar 2022 – Mar 2023)
- IAUTMU Student Research Committee (Feb 2018 – Jun 2024)